



MATHEMATICS (EXTENSION 2)

2013 HSC Course Assessment Task 2

March 18, 2013

General instructions

- Working time – 1 hour.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 5)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M4A – Mr Fletcher

☐ 12M4B – Mr Lam

☐ 12M4C – Ms Ziazaris

Marker's use only.

QUESTION	1-4	5	6	Total	%
MARKS	$\overline{4}$	$\overline{18}$	$\overline{18}$	$\overline{40}$	

Section I: Objective response

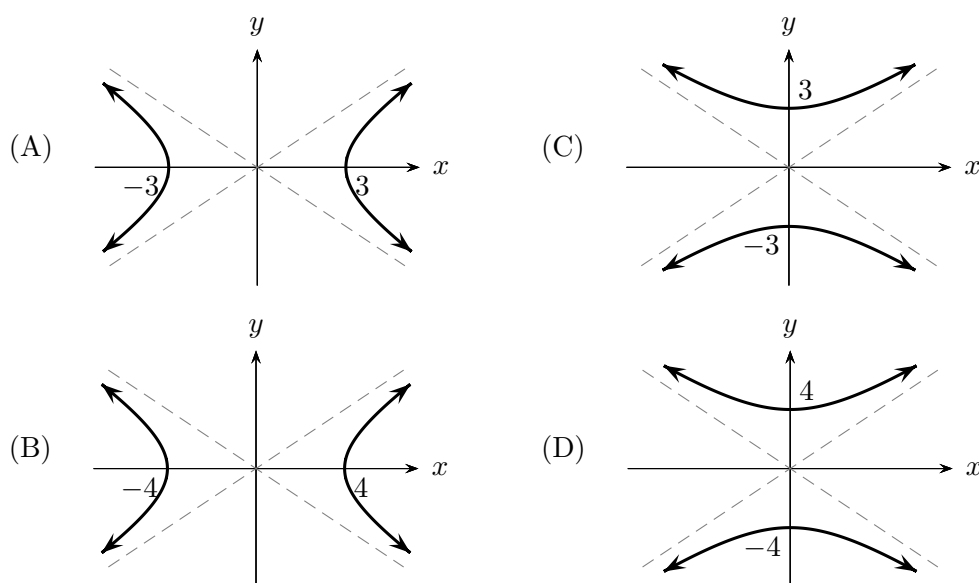
Mark your answers on the multiple choice sheet provided.

Marks

1. $P(z)$ is a polynomial in z of degree 4 with real coefficients. Which one of the following statements must be false? 1

- (A) $P(z)$ has four real roots.
 (B) $P(z)$ has two real roots and two non-real roots.
 (C) $P(z)$ has one real root and three non-real roots.
 (D) $P(z)$ has no real roots.

2. Which of the following is the graph of $16y^2 - 9x^2 = 144$? 1



3. The equation $x^3 + 2x + 1 = 0$ has roots α , β and γ . Which of the following equations has roots 2α , 2β and 2γ ? 1

- (A) $x^3 + 8x + 8 = 0$ (C) $2x^3 + 4x + 2 = 0$
 (B) $x^3 + 16x + 8 = 0$ (D) $8x^3 + 4x + 1 = 0$

4. The points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The chord PQ subtends a right angle at $(0, 0)$. Which of the following is the correct expression? 1

- (A) $\tan \theta \tan \phi = -\frac{b^2}{a^2}$ (C) $\tan \theta \tan \phi = \frac{b^2}{a^2}$
 (B) $\tan \theta \tan \phi = -\frac{a^2}{b^2}$ (D) $\tan \theta \tan \phi = \frac{a^2}{b^2}$

End of Section I
Examination continues overleaf...

Section II: Short answer

Question 5 (18 Marks)

Commence a NEW page.

Marks

- (a) The equation of a hyperbola is given by $4x^2 - 9y^2 = 36$.
- Find the coordinates of S and S' , the foci of the hyperbola. **2**
 - Find the equations of the directrices M and M' . **1**
 - Find the equations of the asymptotes. **1**
 - Let P be any point on the hyperbola. Show that $|PS - PS'| = 6$. **2**

- (b) The line $x = 8$ is the directrix, and $(2, 0)$ is the corresponding focus of the ellipse with equation

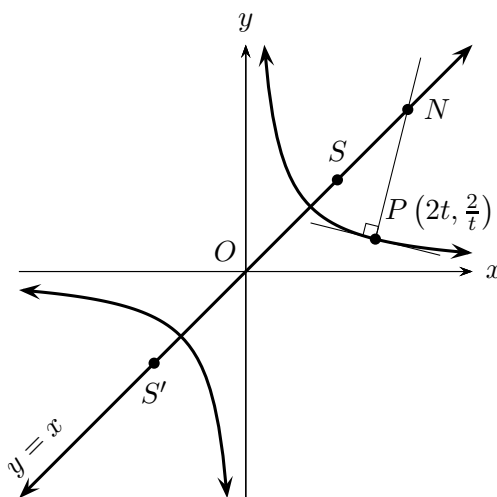
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad a > b > 0$$

Find the value of a and b .

- (c) The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the ellipse meets the x axis at the points A and A' .
- Prove that the tangent at P has the equation **2**

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

- The tangent at P meets the tangents from A and A' at points Q and Q' respectively. Find the coordinates of Q and Q' . **2**
 - The points A , A' , Q' and Q form a trapezium. Prove that the product of lengths of the parallel lines is independent of the position of P . **2**
- (d) The diagram shows the hyperbola $xy = 4$.



- What are the coordinates of the foci S and S' ? **1**

Question 5 continues overleaf...

Question 5 continued from previous page...

The point $P\left(2t, \frac{2}{t}\right)$ lies on the curve, where $t \neq 0$. The normal at P intersects the straight line $y = x$ at N . O is the origin.

Given the equation of the normal at P is

$$y = t^2x + \frac{2}{t} - 8$$

- ii. Find the coordinates of N . **1**
- iii. Show that $\triangle OPN$ is isosceles. **2**

Question 6 (18 Marks)

Commence a NEW page.

Marks

- (a) The polynomial equation $x^3 - 6x^2 + 3x - 2 = 0$ has roots α , β and γ . **2**

Evaluate $\alpha^3 + \beta^3 + \gamma^3$.

- (b) $x^3 + px^2 + qx + r = 0$ has roots α , β and γ , where $\alpha = \beta + \gamma$. **3**

Show that $p^3 - 4pq + 8r = 0$.

- (c) Given $-2 - i$ is a zero of $P(x) = x^4 + 6x^3 + 14x^2 + 14x + 5$, find all zeros of $P(x)$. **3**

- (d) i. The polynomial equation $P(x) = 0$ has a double root at $x = \alpha$. **2**

Show that $x = \alpha$ is also a root of the equation $P'(x) = 0$.

- ii. $y = mx$ is a tangent to the curve $y = 3 - \frac{1}{x^2}$. Show that the equation $mx^3 - 3x^2 + 1 = 0$ has a double root. **1**

- iii. Hence find the equation of any such tangents. **3**

- (e) Given that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$,

- i. Find the roots of $16x^5 - 20x^3 + 5x = 0$. **2**

- ii. Hence show that **2**

$$\cos^2 \frac{\pi}{10} + \cos^2 \frac{3\pi}{10} = \frac{5}{4}$$

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M4A – Mr Fletcher

☐ 12M4B – Mr Lam

☐ 12M4C – Ms Ziazaris

1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

Suggested Solutions

Section I

1. (C) 2. (C) 3. (A) 4. (B)

Section II

Question 5 (Fletcher)

- (a) i. (2 marks)

- ✓ [1] for correct eccentricity.
✓ [1] for correct foci.

$$\begin{aligned} 4x^2 - 9y^2 &= 36 \\ \frac{x^2}{9} - \frac{y^2}{4} &= 1 \\ \therefore a &= 3 \quad b = 2 \end{aligned}$$

Finding the eccentricity,

$$\begin{aligned} b^2 &= a^2(e^2 - 1) \\ 4 &= 9(e^2 - 1) \\ e^2 &= 1 + \frac{4}{9} = \frac{13}{9} \\ \therefore e &= \frac{\sqrt{13}}{3} \end{aligned}$$

Coordinates of foci: $S(\pm ae, 0)$

$$S(\sqrt{13}, 0) \quad S'(-\sqrt{13}, 0)$$

- ii. (1 mark)

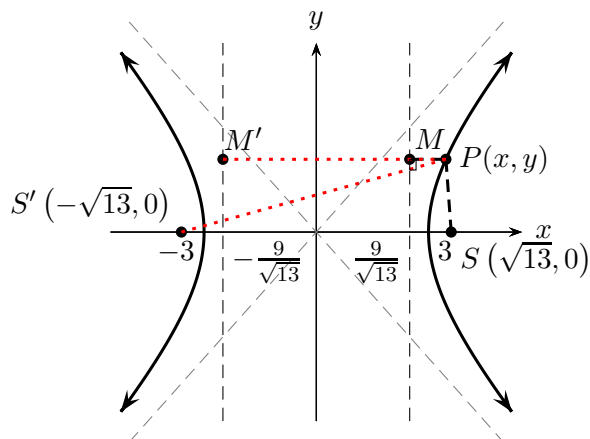
$$x = \pm \frac{a}{e} = \pm \frac{3}{\frac{\sqrt{13}}{3}} = \pm \frac{9}{\sqrt{13}}$$

- iii. (1 mark)

$$y = \pm \frac{b}{a}x = \pm \frac{2}{3}x$$

- iv. (2 marks)

- ✓ [1] for using $PS = ePM$ and $PS' = ePM'$.
✓ [1] for final answer.



$$\begin{aligned} \text{Since } \frac{PS}{PM} &= e \quad \text{and} \quad \frac{PS'}{PM'} = e \\ \therefore \begin{cases} PS = ePM & (1) \\ PS' = ePM' & (2) \end{cases} \end{aligned}$$

Equation (1) subtract equation (2):

$$\begin{aligned} |PS - PS'| &= e |PM - PM'| \\ &= e |MM'| \\ &= \frac{\sqrt{13}}{3} \times 2 \times \frac{9}{\sqrt{13}} \\ &= 6 \end{aligned}$$

- (b) (2 marks)

- ✓ [1] for finding a and e .
✓ [1] for final answer.

$$x = \frac{a}{e} = 8$$

$$(2, 0) = (ae, 0)$$

$$\begin{cases} \frac{a}{e} = 8 & (1) \\ ae = 2 & (2) \end{cases}$$

(1) $\times$ (2):

$$a^2 = 16$$

$$\therefore a = 4$$

$$\therefore e = \frac{1}{2}$$

$$b^2 = a^2(1 - e^2)$$

$$= 16 \left(1 - \frac{1}{4} \right)$$

$$= 16 \times \frac{3}{4} = 12$$

$$\therefore b = 2\sqrt{3}$$

(c) i. (2 marks)

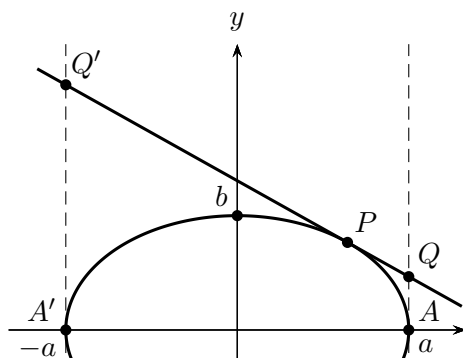
$$\begin{aligned} \begin{cases} x = a \cos \theta \\ y = b \sin \theta \end{cases} \\ \frac{dx}{d\theta} = -a \sin \theta \\ \frac{dy}{d\theta} = b \cos \theta \\ \therefore \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = -\frac{b \cos \theta}{a \sin \theta} \end{aligned}$$

Equation of the tangent at $P(a \cos \theta, b \sin \theta)$ is

$$\begin{aligned} y - b \sin \theta &= -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta) \\ ay \sin \theta - ab \sin^2 \theta &= -bx \cos \theta + ab \cos^2 \theta \\ \underbrace{bx \cos \theta + ay \sin \theta}_{\div ab} &= \underbrace{ab \cos^2 \theta + ab \sin^2 \theta}_{\div ab} \\ \therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} &= 1 \end{aligned}$$

ii. (2 marks)

- ✓ [1] for coordinates of Q
- ✓ [1] for coordinates of Q'



As the equation of the tangent is

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

The coordinates of Q : when $x = a$,

$$\begin{aligned} \frac{a \cos \theta}{a} + \frac{y \sin \theta}{b} &= 1 \\ y \frac{\sin \theta}{b} &= 1 - \cos \theta \\ y &= \frac{b(1 - \cos \theta)}{\sin \theta} \end{aligned}$$

Similarly, the coordinates of Q' :
when $x = -a$

$$\begin{aligned} y &= \frac{b(1 + \cos \theta)}{\sin \theta} \\ \therefore Q &\left(a, \frac{b(1 - \cos \theta)}{\sin \theta} \right) \\ Q' &\left(-a, \frac{b(1 + \cos \theta)}{\sin \theta} \right) \end{aligned}$$

iii. (2 marks)

- ✓ [1] for identifying parallel lines in trapezium and multiplying their lengths.

- ✓ [1] for final answer.

Parallel lines are AQ and AQ' .
Hence

$$\begin{aligned} AQ \times AQ' &= \frac{b(1 - \cos \theta)}{\sin \theta} \times \frac{b(1 + \cos \theta)}{\sin \theta} \\ &= \frac{b^2(1 - \cos^2 \theta)}{\sin^2 \theta} \\ &= b^2 \end{aligned}$$

(d) i. (1 mark)

$$\begin{aligned} xy &= 4 = c^2 \\ \therefore c &= 2 \end{aligned}$$

As $e = \sqrt{2}$, and $S(\pm c\sqrt{2}, \pm c\sqrt{2})$,

$$\therefore S(2\sqrt{2}, 2\sqrt{2}) \quad S'(-2\sqrt{2}, -2\sqrt{2})$$

ii. (1 mark)

$$y = t^2 x + \frac{2}{t} - 8$$

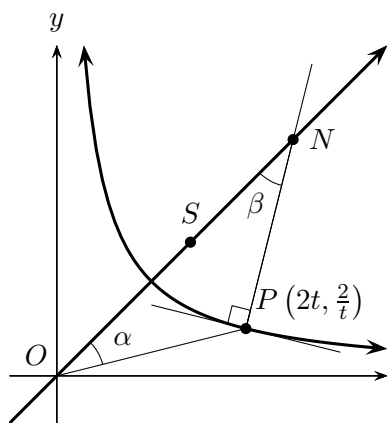
Coordinates of N occur when $y = x$,
i.e.

$$\begin{aligned} x &= t^2 x + \frac{2}{t} - 8 \\ x(1 - t^2) &= \frac{2}{t} - 8 \\ x = y &= \frac{8t - 2}{t(t^2 - 1)} \end{aligned}$$

$$\therefore N \left(\frac{8t - 2}{t(t^2 - 1)}, \frac{8t - 2}{t(t^2 - 1)} \right)$$

iii. (2 marks)

- ✓ [1] for substantial progress towards proof.
- ✓ [1] for correct conclusion.



$$m_{OP} = \frac{\frac{2}{t}}{2t} = \frac{1}{t^2}$$

Gradient of normal:

$$m_{PM} = t^2$$

Angle between ON and OP (α):

$$\begin{aligned}\tan \alpha &= \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| \\ &= \left| \frac{1 - \frac{1}{t^2}}{1 + 1 \left(\frac{1}{t^2} \right)} \right| \times \frac{t^2}{t^2} \\ &= \left| \frac{t^2 - 1}{t^2 + 1} \right|\end{aligned}$$

Angle between ON and PN (β):

$$\tan \beta = \left| \frac{1 - t^2}{1 + 1(t^2)} \right| = \left| \frac{1 - t^2}{1 + t^2} \right|$$

As $\tan \alpha = \tan \beta$ and $\alpha, \beta < 90^\circ$,

$$\therefore \alpha = \beta$$

and $\triangle OPN$ is isosceles.

Question 6 (Fletcher)

(a) (2 marks)

✓ [1] for $\alpha^3 + \beta^3 + \gamma^3$ in terms of α, β and γ .

✓ [1] for final answer.

If α, β and γ are the roots,

$$\alpha^3 - 6\alpha^2 + 3\alpha - 2 = 0$$

$$\beta^3 - 6\beta^2 + 3\beta - 2 = 0$$

$$\gamma^3 - 6\gamma^2 + 3\gamma - 2 = 0$$

Adding and moving terms to other side,

$$\begin{aligned}\alpha^3 + \beta^3 + \gamma^3 &= 6(\alpha^2 + \beta^2 + \gamma^2) - 3(\alpha + \beta + \gamma) + 6\end{aligned}$$

Now to find $\alpha^2 + \beta^2 + \gamma^2$:

$$\begin{aligned}(\alpha + \beta + \gamma)^2 &= (\alpha + \beta)^2 + 2(\alpha + \beta)\gamma + \gamma^2 \\ &= \alpha^2 + 2\alpha\beta + \beta^2 + 2\alpha\gamma + 2\beta\gamma + \gamma^2 \\ \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ \therefore \alpha^3 + \beta^3 + \gamma^3 &= 6 \left[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \right] \\ &\quad - 3(\alpha + \beta + \gamma) + 6 \\ &= 6 \left[(-6)^2 - 2(3) \right] - 3(-6) + 6 \\ &= 168\end{aligned}$$

(b) (3 marks)

✓ [2] (max) for significant progress towards final answer.

✓ [1] for final answer.

$$x^3 + px^2 + qx + r = 0$$

with roots α, β, γ and $\alpha = \beta + \gamma$.

• Using the sum of roots,

$$\begin{aligned}\alpha + \beta + \gamma &= 2\alpha = \frac{-p}{1} \\ \therefore \alpha &= -\frac{p}{2}\end{aligned}$$

• Using the pairs of roots,

$$\begin{aligned}\alpha\beta + \alpha\gamma + \beta\gamma &= \alpha(\beta + \gamma) + \beta\gamma \\ &= \alpha^2 + \beta\gamma \\ &= \frac{p^2}{4} + \beta\gamma = \frac{q}{1} \\ \therefore \frac{p^2}{4} + \beta\gamma &= q\end{aligned}$$

• Using the product of roots,

$$\begin{aligned}\alpha\beta\gamma &= \left(-\frac{p}{2} \right) \left(q - \frac{p^2}{4} \right) \\ &= -\frac{p}{2} \left(q - \frac{p^2}{4} \right) = -r \\ \therefore -\frac{p}{2} \left(q - \frac{p^2}{4} \right) &= -r \\ \underbrace{-\frac{pq}{2} + \frac{p^3}{8}}_{\times 8} &= -r \\ \therefore p^3 - 4pq + 8r &= 0\end{aligned}$$

(c) (3 marks)

- ✓ [1] for identifying conjugate root.
- ✓ [1] for other significant progress towards answer.
- ✓ [1] for final answer.

If $-2 - i$ is a zero to $P(x) = x^4 + 6x^3 + 14x^2 + 14x + 5$, then $-2 + i$ is also a zero.

- Sum of roots:

$$(-2 - i) + (-2 + i) + \alpha + \beta = -\frac{b}{a} = -6$$

$$\alpha + \beta = -2$$

- Product of roots:

$$(-2 - i)(-2 + i) \times \alpha\beta = \frac{c}{a} = 5$$

$$5\alpha\beta = 5$$

$$\alpha\beta = 1$$

$$\begin{cases} \alpha + \beta = -2 \\ \alpha\beta = 1 \end{cases}$$

$$\therefore \alpha = -1 \quad \beta = -1$$

Roots are $-1, -1, -2 - i, -2 + i$.

(d) i. (2 marks)

- ✓ [1] correctly differentiating.
- ✓ [1] for final proof.

Let $P(x) = (x - \alpha)^2 Q(x)$

$$u = (x - \alpha)^2 \quad v = Q(x)$$

$$u' = 2(x - \alpha) \quad v' = Q'(x)$$

$$P'(x) = uv' + vu'$$

$$= (x - \alpha)^2 Q'(x) + 2(x - \alpha)Q(x)$$

$$= (x - \alpha)((x - \alpha)Q'(x) + Q(x))$$

$$= (x - \alpha)Q_3(x)$$

Hence $x = \alpha$ is also a root of $P'(x) = 0$.

ii. (1 mark)

If $y = mx$ is a tangent to $y = 3 - \frac{1}{x^2}$, then

$$mx = 3 - \frac{1}{x^2}$$

has a double root.

$$mx^3 = 3x^2 - 1$$

$$mx^3 - 3x^2 + 1 = 0$$

Hence $mx^3 - 3x^2 + 1 = 0$ has a double root.

iii. (3 marks)

If $P(x)$ has a double root at $x = \alpha$, then $x = \alpha$ is also a root to $P'(x)$:

$$P'(x) = 3mx^2 - 6x$$

$$P'(\alpha) = 3m\alpha^2 - 6\alpha = 0 \Rightarrow 3\alpha(m\alpha - 2) = 0$$

$$\therefore \alpha = 0, \frac{2}{m}$$

As $P'(0) \neq 0$, $x = \frac{2}{m}$ is the double root.

$$P\left(\frac{2}{m}\right) = 0$$

$$\therefore m\left(\frac{2}{m}\right)^3 - 3\left(\frac{2}{m}\right)^2 + 1 = 0$$

$$\frac{8}{m^2} - \frac{12}{m^2} + 1 = 0$$

$$\frac{4}{m^2} = 1$$

$$\therefore m = \pm 2$$

$$\therefore y = \pm 2x$$

(e) i. (2 marks)

Given $\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$, then letting $x = \cos \theta$ will result in the polynomial

$$16x^5 - 20x^3 + 5x = 0$$

Hence to solve the polynomial, solve $\cos 5\theta = 0$:

$$\cos 5\theta = 0$$

$$5\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}$$

$$\therefore \theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}$$

$$\therefore x = 0, \pm \cos \frac{\pi}{10}, \pm \cos \frac{3\pi}{10}$$

ii. (2 marks) Using pairs of roots (the only terms remaining are the ones which do not involve the zero)

$$16x^5 - 20x^3 + 5x = 0$$

$$\left(\cos \frac{\pi}{10}\right) \left(-\cos \frac{\pi}{10}\right)$$

$$+ \left(\cos \frac{3\pi}{10}\right) \left(-\cos \frac{3\pi}{10}\right)$$

$$= -\cos^2 \frac{\pi}{10} - \cos^2 \frac{3\pi}{10} = \frac{c}{a} = -\frac{20}{16}$$

$$\therefore \cos^2 \frac{\pi}{10} + \cos^2 \frac{3\pi}{10} = \frac{5}{4}$$